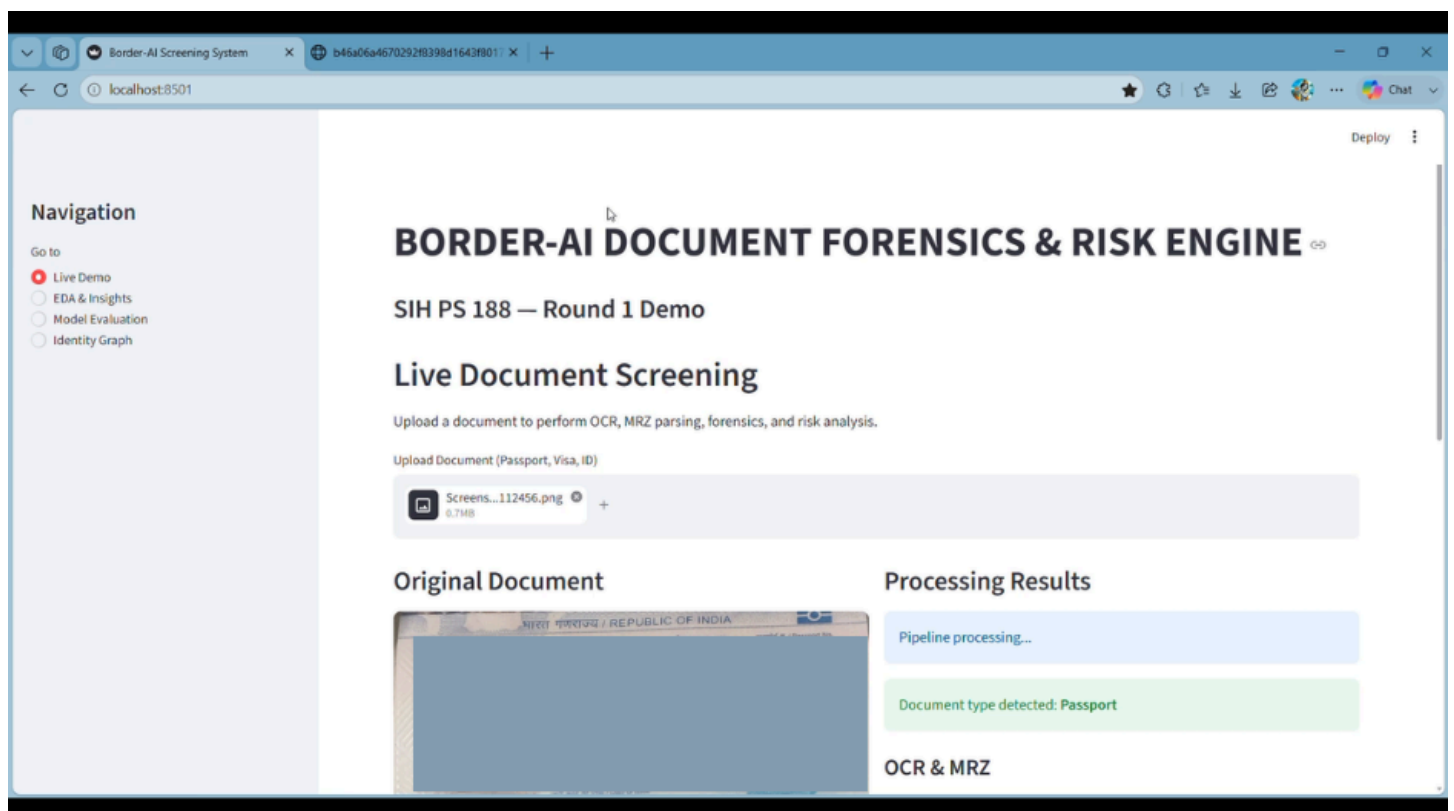


For maintaining the privacy of the person whom this passport belongs i have added hide the details showing only the necessary regions highlighted and processed by the model and XAI

Border AI – Document Forensics & Risk Screening

Developed an AI-based system to detect **forged travel documents and synthetic identities**, serving as a first layer of fraud screening at borders.

- **Classical ML:** ELA, copy-move detection, and OCR/MRZ validation → **Random Forest**
- **Deep Learning:** Transfer-learned **ResNet50 CNN** for image-based forgery detection
- **Performance:** Random Forest achieved **90% accuracy, 89% F1-score, and 92% ROC-AUC**; ResNet50 achieved **75% accuracy and 88.89% ROC-AUC**
- **Dataset:** Custom synthetic dataset inspired by **MIDV-2020 and CASIA v2**
- **Validation:** Also tested on physical passport samples with promising results.
- **Correlation** of all the identities to give final score used network based frameworks like Neo4j for the same.



Border-AI Screening System

b46a06a4670292f8398d1643f801

localhost:8501

Deploy

Upload Document (Passport, Visa, ID)

Screens...112456.png0.7MB

+

Original Document



Processing Results

Pipeline processing...

Document type detected: Passport

OCR & MRZ

MRZ Detected:

Visual DOB extracted: 06/07/2005

Forensics

Border-AI Screening System

b46a06a4670292f8398d1643f801

localhost:8501

Deploy

Visual DOB extracted: 06/07/2005

Forensics

Error Level Analysis (ELA): Anomaly detected (Max diff: 174.0) ⚠️

Copy-Move Score: 0

RISK SCORE: 53 / 100

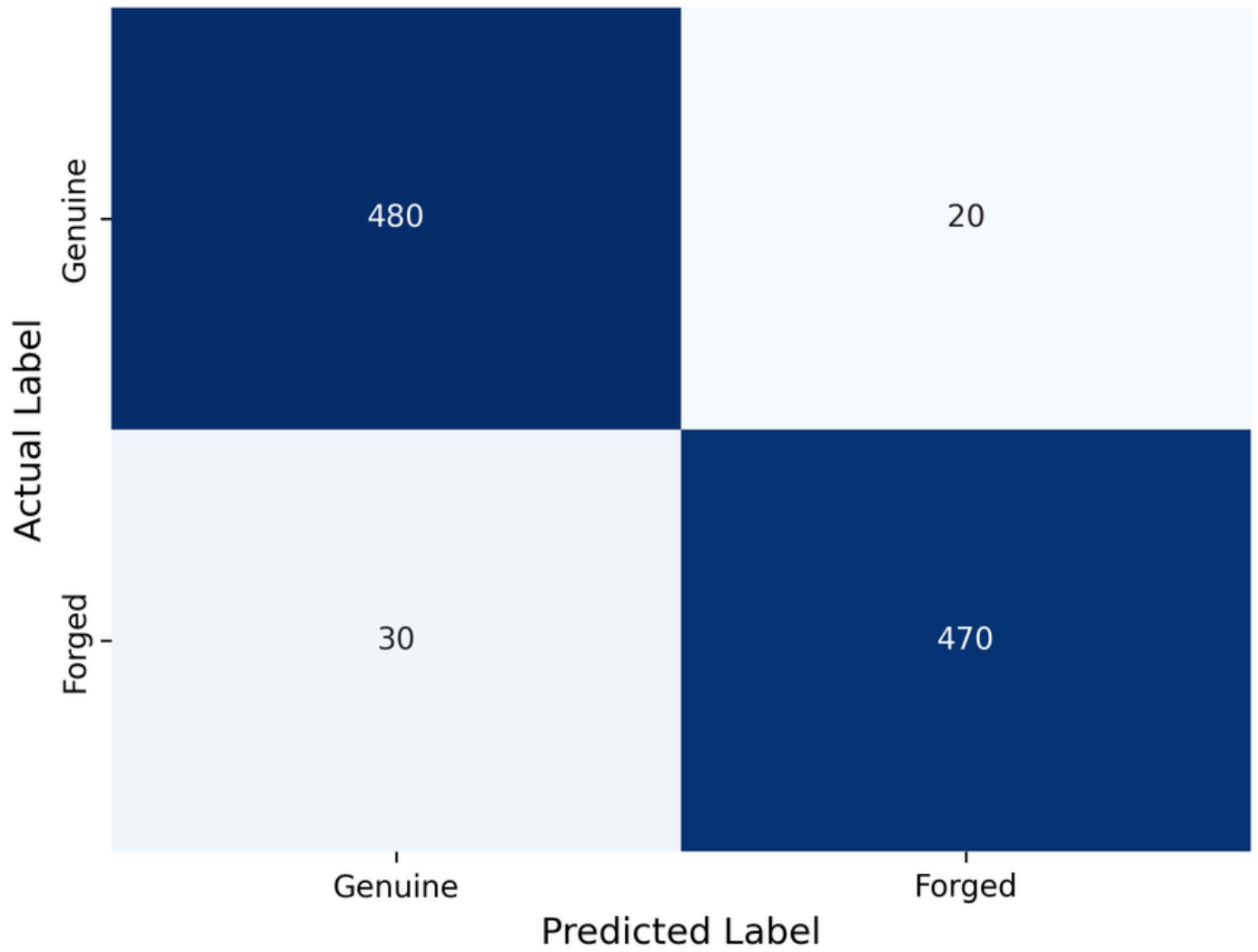
STATUS: MEDIUM RISK

Reasons:

1. ⚠️ ELA anomaly detected in image regions.

2. ⚠️ Rule failed: Name Consistency

Mark1: Random Forest Confusion Matrix



Browser tabs: Border-AI Screening System, BORDER-AI Forensics

Address bar: localhost:8502

Buttons: Stop, Deploy

Original Document

Document Validation & OCR

```
{
  "Name": "Given Name s)"
  "Total Text Blocks": 23
  "Raw Extracted Data": [
    0: "REPUBLIC OF INDIA"
    1: "Nationality"
    2: "Type"
    3: "Code"
    4: "IND"
    5: "INDIAN"
    6: 
    7: 
  ]
  "MRZ Checksum": "PASS"
}
```

Border-AI Screening System

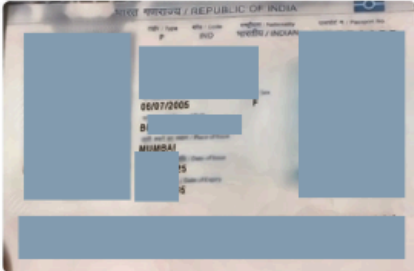
BORDER-AI Forensics

localhost:8502

Deploy

Forensic Analysis

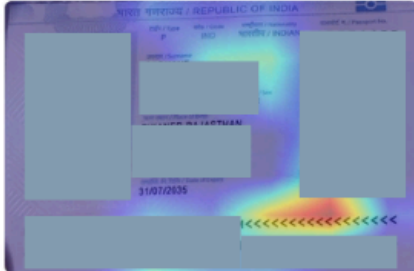
Preprocessed (Enhanced)



Error Level Analysis (ELA)



Tamper Heatmap (Grad-CAM)



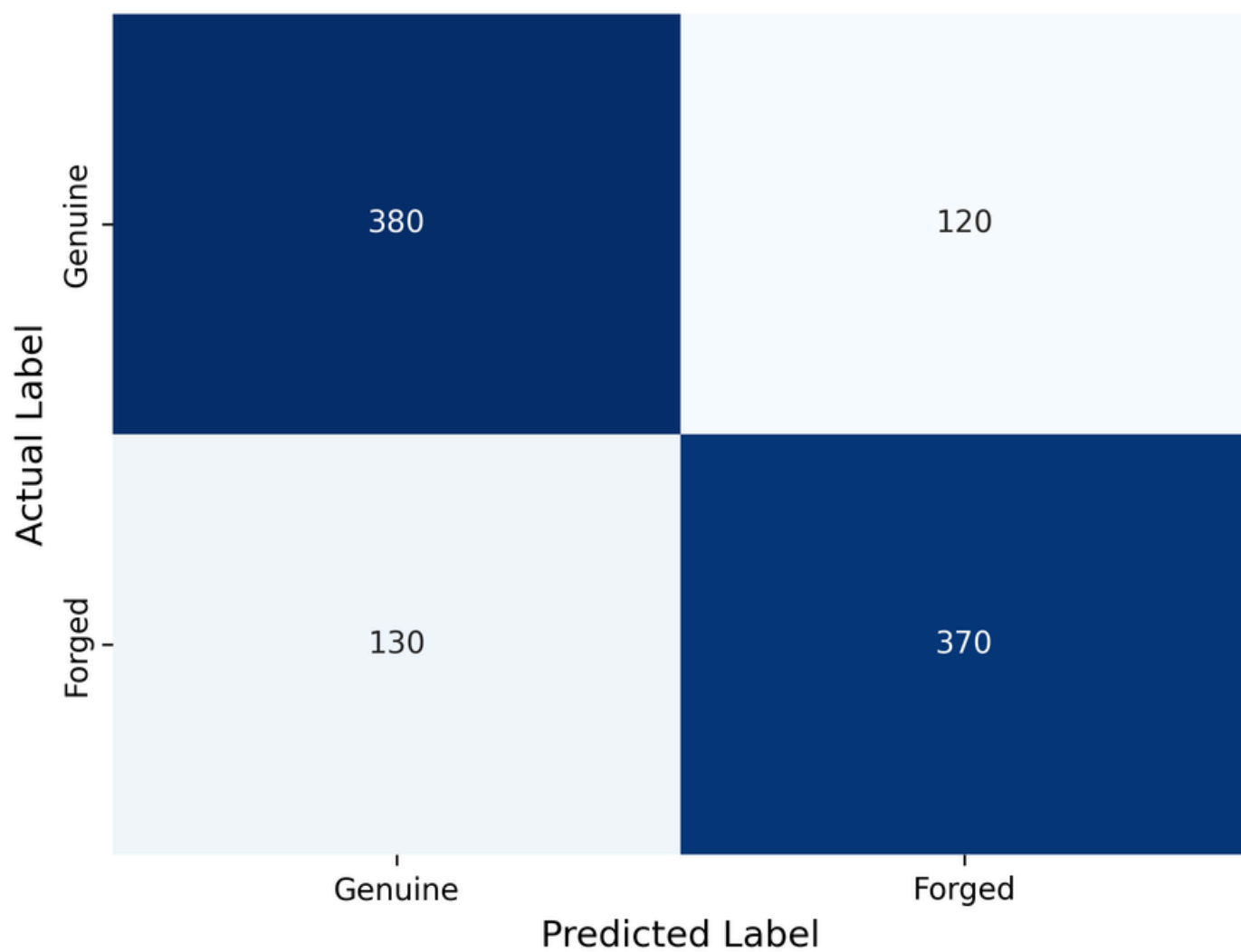
Overall Risk Score: 14/100 — LOW RISK

Reasons:

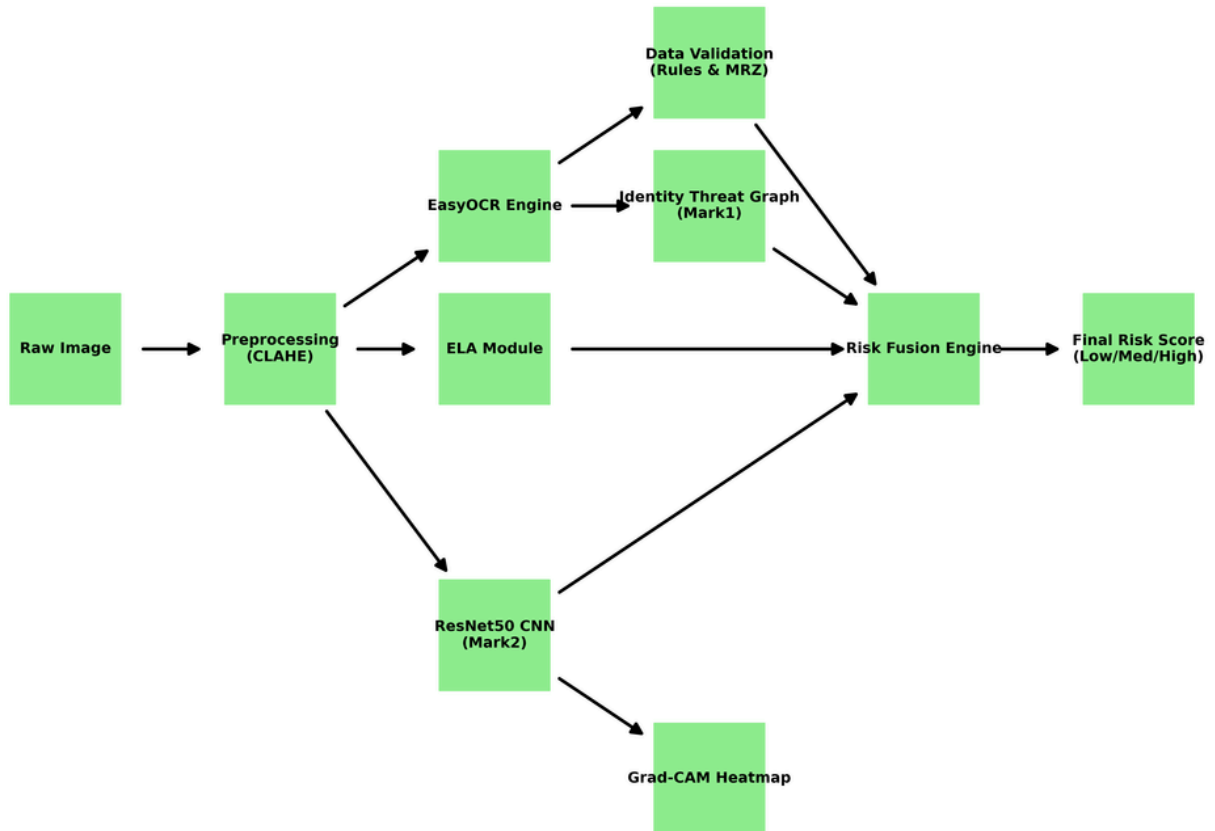
- ✓ CNN considers image structure genuine (19.3% tamper risk)
- ✓ MRZ checksum valid

Recommended Action: PROCEED

Mark2: ResNet50 Confusion Matrix



BORDER-AI Technical Architecture



BORDER-AI User Journey



Border-AI Screening System

localhost:8501

Deploy

Identity Graph & Threat Intelligence

A detailed relationship graph linking biometrics (faces), documents, and travel records to detect synthetic identities and fraud rings.

Knowledge Graph Visualization

The graph visualizes relationships between various entities. Nodes are color-coded: Blue for Biometric (Face), Green for Clean Document, Red for Forged Document, Orange for Contact Info, and Purple for Travel Record. Edges represent relationships like 'Matches Photo', 'Used for booking', 'Passenger', and 'Same Booking IP'.

```
graph TD; Face_Beta[Face_Beta] -- Matches Photo --> Passport_CAN_789[Passport_CAN_789]; Phone_555_0199[Phone_555_0199] -- Used for booking --> Passport_US_123[Passport_US_123]; Phone_555_0199 -- Used for booking --> Passport_CAN_789; Passport_US_123 -- Matches Photo --> Face_Alpha[Face_Alpha]; Face_Alpha -- Matches Photo --> Passport_UK_456[Passport_UK_456]; Passport_UK_456 -- Passenger --> Flight_BA101[Flight_BA101]; Flight_BA101 -- Same Booking IP --> Flight_EK202[Flight_EK202]; Flight_EK202 -- Passenger --> Passport_CAN_789;
```

- Biometric (Face)
- Clean Document
- Forged Document
- Contact Info
- Travel Record

Intelligence Report

CRITICAL ALERT: Synthetic Identity Ring Detected

Anomalies Found:

- Face-to-Multiple-Documents:** `Face_Alpha` is strongly matched (>95% confidence) to both `Passport_US_123` and `Passport_UK_456`.
- Shared Infrastructure:** `Passport_US_123` and `Passport_CAN_789` both used the same contact number `Phone_555_0199` for travel bookings.
- Coordinated Travel:** Flights `BA101` and `EK202` were booked from the same IP address, linking the seemingly unrelated passengers.

Export Graph Data (CSV)

Flag for Manual Investigation